

2016.0 RANGE ROVER (LG), 501-09

REAR VIEW MIRRORS

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For information on the description and operation of the mirrors, refer to the relevant section of the workshop manual. REFER to: [Rear View Mirrors](#) (501-09 Rear View Mirrors, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the warranty policy and procedures manual (section B1.2), or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.

1. Verify the customer concern

2. Visually inspect for obvious mechanical or electrical faults

Visual inspection

MECHANICAL	ELECTRICAL
▪ Door mirror switch condition and installation ▪ Door mirror condition and installation	▪ Battery condition and state of charge ▪ Ambient air temperature sensor resistance check ▪ Fuses ▪ Harnesses and connectors ▪ Washer jet and mirror heater relay ▪ Memory control module(s) ▪ Door mirror switch(es) ▪ Door mirror motor(s) ▪ Ignition switch ▪ Battery junction box ▪ Body control module ▪ Automatic temperature control module ▪ LIN circuit(s)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step
4. Use the approved diagnostic system or a scan tool to retrieve any DTCs before moving onto the symptom chart or DTC index
 - Because the DTCs are stored in more than one module, a complete vehicle read is recommended
 - Make sure that all DTCs are cleared following rectification

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Ambient air temperature sensor intermittent/inoperative/erratic readings or a default setting is appearing (ie -40°). Malfunction indicator lamp lighting or air conditioning is inoperative. (DTC's that might show are: P0070-17, P0071-21, P0071-22, P0071-24, P0071-29, P0073-00, P0072-00, P007B-08, P18A5-14, P1888-15 may also be flagged for this failure.)	▪ Circuit fault: high resistance ▪ Circuit fault: short circuit to ground	▪ Refer to the electrical circuit diagrams and check the ambient air temperature sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. For ambient air temperature resistance check - see below resistance specifications ▪ If the resistance shows to be within the spec stated below, the fault is NOT with the ambient air temperature sensor, and the mirror NOT be replaced
Mirrors do not defrost/washer jets freeze	▪ Fuse fault ▪ Washer jet and mirror heater relay fault ▪ Circuit fault: high	Check the fuses. Check the operation of the washer jet and mirror heater relay (see the visual inspection). Check the washer jet and mirror heater circuits. Refer to the electrical guides

	- resistance ■ Circuit fault: short circuit to ground ■ Automatic temperature control module fault	
Mirrors inoperative in one or more directions	■ Mechanical fault ■ Switch fault ■ Motor fault ■ Circuit fault: high resistance ■ Circuit fault: short circuit to ground	Operate the mirror switch and listen for the motor(s). If the motor(s) can be heard, check the mechanical condition of the mirror and linkages. Rectify as necessary. Check for DTCs indicating a switch, motor or circuit fault
Memorized mirror position is not resumed	■ Battery voltage below 10.5 volts ■ Position not stored ■ Switch operated during "one-touch" memory recall ■ EEPROM fault	Before condemning a memory component, check the function from the switch and refer to the symptoms above. Make sure the battery voltage is adequate. Make sure that the desired position is correctly stored. Make sure that the memory store/recall procedure is being followed. Refer to the relevant section of the workshop manual
'Lazy entry' function inoperative	■ Remote transmitter fault (battery, transmitter programming, etc) ■ See list for "position is not resumed"	Check that the remote transmitter operates the central locking, etc. If it does, the fault is not with the transmitter. Refer to symptom "mirror position is not resumed" (above)

PROCEDURE

When door mirrors fitted with an ambient air temperature sensor are removed and reconnected or replaced, the ambient air temperature sensor will initially give a false external temperature reading of -40°C at the next start-up. The sensor will need to be correctly calibrated in order for the air conditioning to function correctly.

In order for the temperature sensor to be correctly configured to the actual temperature, using Jaguar Land Rover approved diagnostic equipment, run the routine - Reset/Clear Specified Function (0x040E) from the automatic temperature control module using the command 31 01 04 0E 17 (Reset Ambient Air Temperature Algorithm), then run the command 11 40

AMBIENT AIR TEMPERATURE SENSOR SPECIFICATIONS

TEMPERATURE (°C)	R NOMINAL (OHM)	R MINIMUM (OHM)	R MAXIMUM (OHM)	RESISTANCE TOLERANCE (± %)	TEMPERATURE TOLERANCE (± °C)
-40	65319.0308	62679.33	67958.73	4.0	0.6
-35	46990.0095	45205.83	48774.19	3.8	0.6
-30	34207.5960	32989.33	35425.86	3.6	0.6
-25	25182.1798	24342.59	26021.77	3.3	0.6
-20	18734.6254	18151.13	19318.13	3.1	0.5
-15	14077.5400	13668.95	14486.13	2.9	0.5
-10	10678.4286	10390.41	10966.45	2.7	0.5
-5	8172.8415	7968.63	8377.05	2.5	0.5
0	6308.5000	6163.00	6454.00	2.3	0.5
5	4909.3071	4786.93	5031.68	2.5	0.5
10	3849.8711	3746.95	3952.79	2.7	0.6
15	3041.2012	2954.58	3127.83	2.8	0.6
20	2419.1836	2346.16	2492.20	3.0	0.7
25	1937.2187	1875.55	1998.89	3.2	0.7
30	1561.1510	1508.95	1613.35	3.3	0.8

35	1265.7445	1221.45	1310.04	3.5	0.8
40	1032.2097	994.53	1069.89	3.7	0.9
45	846.4540	814.31	878.60	3.8	1.0
50	697.8327	670.33	725.34	3.9	1.0
55	578.2514	554.65	601.85	4.1	1.1
60	481.5150	461.21	501.82	4.2	1.2
65	402.8524	385.33	420.37	4.3	1.2
70	338.5673	323.41	353.73	4.5	1.3
75	285.7796	272.62	298.94	4.6	1.4
80	242.2327	230.78	253.68	4.7	1.4
85	206.1492	196.16	216.14	4.8	1.5

DTC INDEX

For a complete list of all diagnostic trouble codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: (100-00 General Information)

[Diagnostic Trouble Code Index - DTC: Driver/Passenger Door Module \(DDM/PDM\)](#) (Description and Operation),

[Diagnostic Trouble Code Index - DTC: Body Control Module \(BCM\)](#) (Description and Operation).